

MEMORANDUM TO THE FILE

Subj: USE OF OTHER THAN FULL AND OPEN COMPETITION FOR AN ACQUISITION UNDER THE SIMPLIFIED ACQUISITION THRESHOLD

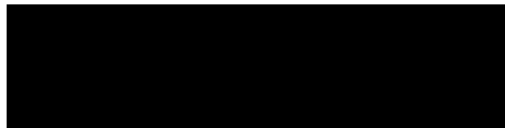
IDENTIFICATION OF SOURCE:

Vendor Name: TA Instrumets - Waters, LLC

Vendor City/State: Newcastle, DE

Cage: 0MJSS

Total Dollar Value:



1. FAR 13.106-1(b)(1)(i) states for purchases not exceeding the simplified acquisition threshold, Contracting Officers may solicit from one source if the Contracting Officer determines that the circumstances of the contract action deem only one source reasonably available (e.g., urgency, exclusive licensing agreements, brand-name or industrial mobilization).

2. As required by FAR 13.106-3(b)(3)(i), this memorandum explains the absence of competition for this acquisition, as only one source will be solicited for this requirement (or for a portion of the requirement) and the requirement is not expected to exceed the simplified acquisition threshold.

II. DESCRIPTION OF ITEM OR SERVICE:

Provide a description of the item or service required, to include its intended use, the name of the required source or brand name, and the required delivery date/period of performance.

The CHAMP Engineering laboratory supports material evaluation for multiple programs across NSWC Crane, include TR-343, TR-317, SP-22, and NAVAIR. Thermal analysis equipment is required to assess the thermal properties and characteristics of experimental materials to enable identification of heat capacity, material stability, and decomposition processes. The TA Instruments Thermogravimetric Analyzer (Discovery TGA 5500) and Differential Scanning Calorimeter (Discovery DSC 2500) are being procured to assess the thermal properties and characteristics of materials, including high temperature materials for hypersonic applications. TGA and DSC are paired because they provide complementary data on a single sample under identical conditions. Accessories needed include 4-port broadband router for equipment communication, and environmental and sample creation accessories including: Gas Dryer, Press and Dies Kit, Blending Gas Delivery Module, Sample Pan/Bail Ceraminc 100UL, Sample Pan/Bail Ceramic 250UL, Kit Capsule Hi temp Hi-Press DSC, Hangdown Hook Tare TGA, Hangdown Wire Sample TGA, and Kit Halogen Bulbs – Ceramic Wrap. Sole source is being sought because it is critical all equipment and accessories can communicate with each other. TA Instruments have historically been used at Navy laboratories for analysis of similar materials and usage of the same company allows for direct comparison of data between laboratories. TA Instruments is capable of achieving the high temperature and low temperature environments, with the gases needed for analysis, at the precision and sensitivity needed for analysis as detailed in the sole source documentation. Required source is the OEM of the compatible test equipment, TA Instruments. No piece of equipment is compatible with an alternative brand. The required deliver and installation is 60 days from the date of contract award.

III. THE DETERMINATION FOR USE OF OTHER THAN FULL AND OPEN COMPETITION IS BASED ON THE FOLLOWING (select applicable area below to open specific form)

- ☒ A. ONLY ONE RESPONSIBLE SOURCE
- ☒ B. BRAND NAME
- ☐ C. UNUSUAL AND COMPELLING URGENCY

A. ONLY ONE RESPONSIBLE SOURCE (select all that apply):

1. RESTRICTIVE RIGHTS. The source has established proprietary rights, limited rights in data, patent rights, copyrights or secret processes in the item or service required.

2. EXCLUSIVE LICENSING AGREEMENTS. The item or service is only available from the Original Equipment Manufacturer (OEM), or there is only one authorized distributor or technical representative for the OEM.

3. OTHER REASONS. Other reason not identified above. Provide details on mission impact if requirement is not solicited from only one source.

Significant delays in test capabilities and material approvals for critical systems will occur if an TA Instrumets brand thermal analysis test suite is not procured. Comparable materials tested by the Naval Research Laboratory and Air Force Research Laboratory were collected with TA equipment. Significant testing time and training time has already been invested in TA equipment. Historical data will not be comparable if TA Instruments DSC and TGA are not used, and will require time and consumables to re-evaluate the materials and identify equipment biases with a different brand's test system. The DSC, TGA, and accessories also may not be sourced from different brands. Fixtures and accessories are not cross-compatible between different brands and must be purchased as an equipment suite from a single brand.

B. BRAND NAME. Item required must be a brand name product, or feature of a product, particular to one manufacturer.

The features of TA Instruments TGA/DSC test suite meet the temperatures, heat ramp rates, precision, and gas flow rates necessary to complete analysis of the material handled by the GXN CHAMP Laboratory. Because of the highly precise nature of the intended experiments, the instrumentation, particularly the DSC instrument, must meet the following minimum specifications. The materials to be analyzed exhibit weak transitions which require a perfectly flat baseline and excellent baseline repeatability. For this reason, the DSC must exhibit the following minimum baseline performance:

Temperature range: -85°C - 500°C using mechanical cooling and nitrogen purge gas. This specification must be demonstrated by inclusion of real data.

Baseline curvature (-50°C – 300°C): < 20 microW of curvature, as calculated by the maximum deviation from a linear interpolation of the temperature extremes. This baseline performance must also be repeatable and reproducible over a minimum of 10 repeat runs, with no more than 20 microW deviation. This performance must be documented and proven by inclusion of real data.

Baseline noise: < 1 microW @ 1°C/min (without smoothing). This specification must be achievable over the entire operating range of the instrument with a single measurement sensor i.e. it should not be necessary to employ multiple sensors to achieve this specification over the entire temperature range. This specification must be demonstrated by inclusion of real data.

Purge Gas Control: The DSC must include digital mass flow control. These mass-flow controllers must be incorporated into the instrument (i.e. shall not be a separate unit). Flow rate must be programmable within operating software, and deliverable as a saved signal in the data file. Mass flow controllers must also allow for automated gas-switching during an experiment.

Modulated Temperature DSC: GXN CHAMP Laboratory intends to use this instrument to study polymer degradation, amongst other material degradation. For this reason, the DSC must be capable of modulated temperature DSC in both the heating and pseudo-isothermal modes. Pseudo-isothermal is defined as holding an average isothermal temperature with a superimposed temperature modulation. This specification must be documented with real data or published briefs or articles showing data collected in this mode. The modulated temperature profile must be a regular, periodic function of known amplitude and frequency to ensure steady state control during the experiment, as well as experimental reproducibility. Random temperature perturbations are less reproducible and as such are not acceptable for our research. Baseline subtractions should not be required in the MT-DSC mode. Preferably, data deconvolution should be performed in the instrument (not post-experiment), so that heat capacity can be monitored real-time. The following signals should be available from the MT-DSC experiment:

Complex Heat Capacity (C*), In-Phase Cp (C*), Out-of-phase Cp (C*), Phase angle, Heat Flow, Cp (Reversing) Heat Flow, Kinetic (Nonreversing) Heat Flow, Modulated Temperature, Modulated Heat Flow

The performance of the TGA shall be as follows:

Temperature range: ambient-1200°C

Weighing Capacity: 1 gram

Balance Sensitivity: 0.1 microgram

Weighing Precision: 0.01% at 100 mg

Baseline Drift: < 50 microgram (ambient to 1200°C (no baseline subtraction). The TGA should preferably be of vertical null-balance design to minimize baseline drift due to buoyancy and beam growth. This specification should be demonstrated by inclusion of real data.

Purge Gas Control: The TGA must include digital mass flow control. These mass-flow controllers must be incorporated into the instrument (i.e. shall not be a separate unit). Flow rate must be programmable within operating software, and deliverable as a saved signal in the data file. Mass flow controllers must also allow for automated gas-switching during an experiment. The TGA should employ horizontal purging and a short (<1 in.) horizontal exhaust path for potential offgas analysis.

Furnace Cooling: The TGA should employ active cooling (forced-air or other) to efficiently cool the TGA furnace from 1200°C to 35°C in <10 min. This specification should be demonstrated by inclusion of real data.

C. UNUSUAL AND COMPELLING URGENCY. The supplies or services are of such unusual and compelling urgency that the Government would be seriously injured unless the agency is permitted to limit the number of sources from which it solicits, bids, or proposals. Urgent requirements are supplies or services that will result in a work stoppage, or are mission critical and the routine processing time would result in injury to the Government. The determination that the procurement for the above item/service is an urgent and compelling requirement is based on the following:

1. Date on which the requirement was first identified:

2. Required delivery date/period of performance

3. Explanation of why delivery/commencement of services by the date mentioned above is required:

4. If appropriate, include any additional pertinent information.

IV. CERTIFICATION/APPROVALS:

A. Technical / Requirements Certifications: I certify that the facts and representation under my cognizance which are included in this justification are complete and accurate to the best of my knowledge and belief.



Signature

Name (Printed) / Phone #

Date

B. Certification / Approval

Upon the basis of the above statements and information justification, I hereby approve, as the Contracting Officer, the solicitation of the proposed procurement(s) using other than full and open competition.

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Digitally signed by
HELTON.MORGAN.MICHELLE
JUNALEE [REDACTED]
Date: 2026.03.04 09:41:10 -05'00'

Morgan Helton

03/04/2026

Contracting Officer Signature

Name (Printed) / Phone #

Date